

GeoAI LAB

Soil Texture from Spectra

To-do list — Work in order — One deliverable per task

The Data: One row = one soil sample analysed in a laboratory, with its spectrum.

Column	Content
L400 ... L2500	Reflectance, every 5 nm, from 400 to 2500 nm
sand, silt, clay	The three texture fractions — the targets
programme	The survey it comes from: another laboratory, region, instrument
instrument	The device that measured the spectrum
depth_top, depth_bottom	Sampling depth, in cm
lon, lat	Sample location

Rules: Work in order. Produce every deliverable. The method is never given: finding it is the exercise.

1. DATA EXPLORATION

Purpose: Know what you are holding before you model anything. Every decision you make later depends on what you find in this section.

#	Task	Deliverable
1.1	Load the file. Report its size and its columns.	shape + column list
1.2	Check missing values, duplicates and value ranges.	1 summary table
1.3	Plot 20 spectra on the same axes.	1 figure
1.4	Plot the mean spectrum with ± 1 standard deviation.	1 figure
1.5	Plot the mean spectrum, one curve per programme.	1 figure
1.6	Run a PCA on the spectral columns only.	Number of components reaching 95% of variance
1.7	Compute the correlation between two neighbouring wavelengths.	1 number
1.8	Plot the distribution of sand, silt and clay.	1 figure
1.9	Compute sand + silt + clay for every sample.	min, mean, max
1.10	Place the samples in the soil texture triangle.	1 figure
1.11	Count the samples per texture class.	1 table
1.12	Map the samples using lon / lat.	1 figure
1.13	Count samples per programme, and box-plot clay per programme.	1 table + 1 figure
1.14	Plot the histogram of the sampling depth.	1 figure
1.15	For each wavelength, compute its correlation with clay, silt and sand. Plot the three curves against wavelength.	1 figure + the peak wavelength of each target, written down

2. SENSOR SIMULATION

Purpose: Your spectra were measured in a laboratory, on a continuous wavelength grid. A satellite does not measure that way: it returns one number per band. Produce what each of the two satellites below would have returned for these same soils.

#	Task	Deliverable
2.1	Convert the laboratory spectra into the 12 bands defined in the table below.	1 table, 12 columns + 1 sentence stating the rule you used
2.2	Convert them again into bands placed every 12 nm from 400 to 2500 nm, each 12 nm wide.	1 table, ~175 columns
2.3	Plot one sample three times on the same axes: laboratory curve, hyperspectral version, multispectral version.	1 figure
2.4	Plot the same three versions of the mean spectrum, zoomed on 1900–2500 nm.	1 figure
2.5	State and justify the rule you chose in 2.1, and name one alternative you rejected.	3 written lines

Multispectral sensor: The 12 bands

Band	Centre (nm)	Width (nm)	Band	Centre (nm)	Width (nm)
B1	443	20	B7	783	20
B2	490	65	B8	842	115
B3	560	35	B8A	865	20
B4	665	30	B9	945	20
B5	705	15	B11	1610	90
B6	740	15	B12	2190	180

3. BEFORE MODELLING

Purpose: Decide what is achievable before any score can influence you, and put it in writing. No model in this section.

#	Task	Deliverable
3.1	Redo task 1.15 on each of the two simulated datasets.	1 figure per sensor
3.2	For each target and each sensor, state whether the information survived the simulation.	3×2 table (yes/no/partly) + a short reason per cell
3.3	Write down your prediction: which target will be easiest, which hardest, and which loses most when going from hyperspectral to multispectral.	3 written lines (do not modify them later)

4. FIRST MODEL

Purpose: Get a baseline and inspect the predictions themselves, not only the score.

#	Task	Deliverable
4.1	Random train/test split. Train a model per target on the 12-band data.	3 scores
4.2	Plot predicted against measured, for each target.	3 figures
4.3	For each test sample, sum the three predicted fractions.	min, mean, max + % of samples further than 2% from 1
4.4	Plot the histogram of that sum.	1 figure

5. VALIDATION SCENARIOS

Purpose: Measure how the same model behaves when the evaluation conditions change, and when the input representation changes.

#	Task	Deliverable
5.1	Re-evaluate without mixing surveys: train on all programmes but one, test on the one left out. Repeat for each programme.	1 score per programme
5.2	Fill the table below, under both evaluations.	Completed table
5.3	Plot the table as a grouped bar chart.	1 figure
5.4	Compare with what you wrote in 3.3.	1 written line

Evaluation Table Template (Task 5.2)

Configuration	Random split	One programme left out
Clay — Laboratory spectrum		
Clay — Hyperspectral		
Clay — Multispectral		
Silt — Laboratory spectrum		
Silt — Hyperspectral		
Silt — Multispectral		
Sand — Laboratory spectrum		
Sand — Hyperspectral		
Sand — Multispectral		

6. EXPLAINABILITY

Purpose: Find out what the model actually uses — and how far that answer can be trusted.

#	Task	Deliverable
6.1	Take your clay model. Measure which inputs it relies on, with a first method.	1 curve: importance against wavelength
6.2	Do it again with a different method.	1 curve
6.3	Plot the two on the same axes.	1 figure
6.4	Re-run 6.1 three times with three different random seeds.	1 figure with the three curves
6.5	Make the importance profile stable.	1 figure + 1 sentence describing what you did
6.6	From section 5, identify one programme that transfers well and one that transfers badly. Train a model on each, and compare what they rely on.	1 figure with both profiles